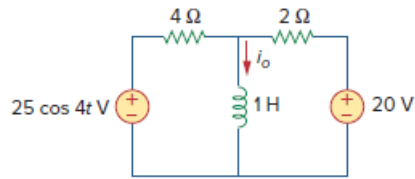


Problem

Find i_o in the circuit shown in Fig. 10.85 using superposition.

Figure 10.85:



Step-by-step solution

Step 1 of 8

Refer to Figure 10.85 in the textbook.

Determine the value of inductive reactance.

$$\begin{aligned} X_L &= j\omega L \\ &= j(4)(1) \\ &= j4 \end{aligned}$$

Re-draw the circuit diagram.

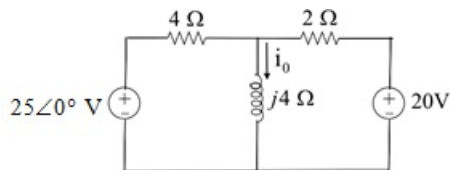


Figure 1

Step 2 of 8

Considering the voltage source $V_1 = 25\angle 0^\circ \text{ V}$ alone and short circuiting the voltage source $V_2 = 20 \text{ V}$, re-draw the circuit diagram.

Let the currents I'_1 , I'_2 and I'_3 flow as shown in figure 2.

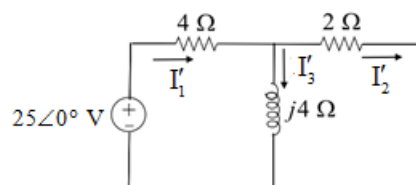


Figure 2

Step 3 of 8

The total impedance of the network to the source V_1 is,

$$\begin{aligned}Z'_1 &= 4 + (2 \parallel j4) \\&= 4 + \frac{2(j4)}{2 + j4} \\&= \frac{8 + j16 + j8}{2 + j4} \\&= \frac{8 + j24}{2 + j4}\end{aligned}$$

$$Z'_1 = 5.6 + j0.8 \, \Omega$$

Step 4 of 8

Calculate the value of current, I'_1 .

$$\begin{aligned}I'_1 &= \frac{V_1}{Z'_1} \\&= \frac{25 \angle 0^\circ}{5.6 + j0.8} \\&= \frac{25 \angle 0^\circ}{5.657 \angle 8.13^\circ} \\&= 4.42 \angle -8.13^\circ \text{ A}\end{aligned}$$

Determine the value of current, I'_3 .

$$\begin{aligned}I'_3 &= I'_1 \left(\frac{2}{2 + j4} \right) \\&= (4.42 \angle -8.13^\circ) \left(\frac{2}{2 + j4} \right) \\&= 1.977 \angle -71.56^\circ \text{ A}\end{aligned}$$

Step 5 of 8

Next considering the voltage source $V_2 = 20\text{ V}$ alone and short circuiting the voltage source $V_1 = 25\angle 0^\circ\text{ V}$, the network reduces to the form shown in figure 3.

Let the currents I_1'' , I_2'' and I_3'' flow as shown in figure 3.

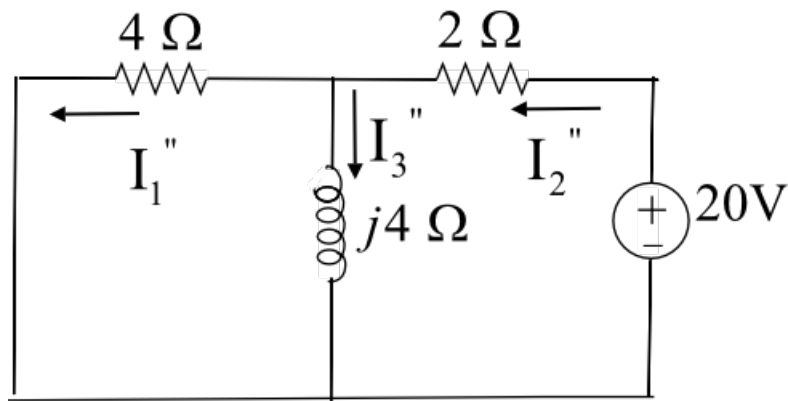


Figure 3

From Figure 3, the source is dc source. Hence, the inductor acts as short-circuit.

Re-draw the circuit diagram.

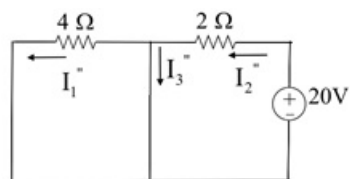


Figure 4

Step 6 of 8

From Figure 4, there is an short-circuit as $4\ \Omega$ resistor. Hence, no current enters into $4\ \Omega$ resistor.

Re-draw the circuit diagram.

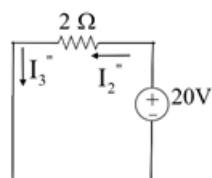


Figure 5

Step 7 of 8

Determine the value of current, I_3'' .

$$\begin{aligned} I_3'' &= \frac{V_2}{2} \\ &= \frac{20}{2} \\ &= 10\text{ A} \end{aligned}$$

Step 8 of 8

When both the voltage sources V_1 and V_2 are considered, then by superposition theorem,

$$\begin{aligned}i_0 &= I'_3 + I''_3 \\&= 1.977 \angle -71.56^\circ + 10 \angle 0^\circ \\&= 10.79 \angle -10.01^\circ \text{ A}\end{aligned}$$

The value of current, i_0 in time domain is,

$$i_0(t) = 10.79 \cos(4t - 10.01^\circ) \text{ A}$$

Therefore, the value of current, i_0 is $\boxed{10.79 \cos(4t - 10.01^\circ) \text{ A}}$.